Persistence of High-Quality Product Use under Forgetting

Standalone theoretical note

This note presents the proofs of the discussed proposition about the frequent use of the high-quality product.

Model and notation. Fix $\rho \in (0,1)$, $\gamma \in (0,1)$, $p_0 \in (0,1)$, $0 < p_1 < p_2 < 1$, and $0 \le c_1 < c_2 < R$. Write $\Delta c = c_2 - c_1$ and $\Delta p = p_2 - p_1$. As in the manuscript, the normalized state space and outcome transitions are

$$\begin{aligned} \mathcal{Z} &= \{(x_s, x_f) : x_s \ge 0,\ x_f \ge 0,\ x_s + x_f \le 1\}, \\ T_\rho^s(x_s, x_f) &= (\rho x_s + 1 - \rho, \rho x_f), \\ T_\rho^f(x_s, x_f) &= (\rho x_s, \rho x_f + 1 - \rho). \end{aligned}$$

Lemma 1 (Continuity of V_ρ)**.** *For every $\rho \in [0,1)$, the Bellman equation has a unique bounded continuous solution V_ρ on $\mathcal{Z}$.*

Proof. Let $C(\mathcal{Z})$ be the space of continuous real-valued functions on $\mathcal{Z}$, with norm $\|v\|_\infty = \sup_{z \in \mathcal{Z}} |v(z)|$. Continuity of demand and the transition maps implies that the Bellman operator $\mathcal{T}_\rho$ maps $C(\mathcal{Z})$ into itself. Since

$$0 \le \frac{D_\rho(z)}{1 - \gamma[1 - D_\rho(z)]} \le 1,$$

we have

$$\|\mathcal{T}_\rho v - \mathcal{T}_\rho w\|_\infty \le \gamma \|v - w\|_\infty.$$

Thus $\mathcal{T}_\rho$ is a contraction. As $\mathcal{Z}$ is compact, $C(\mathcal{Z})$ is complete, and the Banach fixed-point theorem gives a unique bounded continuous solution V_ρ. □

The boundary set. For $0 < \rho < 1$, define

$$\mathcal{K}_\rho = \left\{ \left((1-\rho) \sum_{j=0}^{\infty} \rho^j b_j, (1-\rho) \sum_{j=0}^{\infty} \rho^j (1 - b_j) \right) :\ b_j \in \{0,1\} \text{ for every } j \ge 0 \right\}. \tag{1}$$

Here $b_j = 1$ represents a success and $b_j = 0$ a failure, with $j = 0$ representing the most recent outcome. Since

$$(1-\rho) \sum_{j=0}^{\infty} \rho^j = 1,$$

every point of $\mathcal{K}_\rho$ lies on the boundary segment $x_s + x_f = 1$. Equivalently,

$$\mathcal{K}_\rho = \{(x, 1-x) : x \in C_\rho\}, \qquad C_\rho = \left\{ (1-\rho) \sum_{j=0}^{\infty} \rho^j b_j : b_j \in \{0,1\} \text{ for every } j \ge 0 \right\}. \tag{2}$$

We call $\mathcal{K}_\rho$ the limiting reachable set: its points can be approached arbitrarily closely by states generated from $(0,0)$ through increasingly long histories of successes and failures. It satisfies

$$\mathcal{K}_\rho = T_\rho^s(\mathcal{K}_\rho) \cup T_\rho^f(\mathcal{K}_\rho). \tag{3}$$

If $\rho \geq 1/2$, then $C_\rho = [0,1]$, so $\mathcal{K}_\rho$ is the entire boundary segment. If $0 < \rho < 1/2$, then C_ρ is a Cantor set, so $\mathcal{K}_\rho$ is only a subset of that segment.

Thus $\mathcal{K}_\rho$ describes limiting reachable states, rather than states reached after finitely many observations. In particular, from $(0,0)$ and for $0 < \rho < 1$, the coordinate sum after n observations is $1 - \rho^n < 1$. Such states approach the boundary without reaching it in finite time. We next verify compactness of the limiting set.

Lemma 2 (Compactness of $\mathcal{K}_\rho$)**.** *For $\rho \in (0,1)$ we have that the set $\mathcal{K}_\rho$ is compact.*

Proof. On the boundary, write $(x_s, x_f) = (x, 1-x)$. Start with all possible first coordinates, $K_0 = [0,1]$, then

$$K_{n+1} = f_0(K_n) \cup f_1(K_n), \qquad f_0(x) = \rho x, \quad f_1(x) = \rho x + (1-\rho).$$

Since K_0 is compact and both maps are continuous, induction shows that every K_n is compact. Moreover, $K_1 \subseteq K_0$; applying both maps to each inclusion gives

$$K_{n+1} \subseteq K_n \qquad \text{for every } n \geq 0.$$

We now prove the equality

$$C_\rho = \bigcap_{n=0}^{\infty} K_n.$$

First inclusion: $C_\rho \subseteq \bigcap_{n\geq 0} K_n$. Let $x \in C_\rho$. By definition, there are digits $b_j \in \{0,1\}$ such that $x = (1-\rho)\sum_{j=0}^{\infty} \rho^j b_j$. For any $n \geq 1$, separate the first n terms:

$$x = (1-\rho)\sum_{j=0}^{n-1} \rho^j b_j + \rho^n y_n, \qquad y_n = (1-\rho)\sum_{k=0}^{\infty} \rho^k b_{n+k} \in [0,1].$$

This is precisely the result of applying n transitions to y_n:

$$x = f_{b_0} \circ f_{b_1} \circ \cdots \circ f_{b_{n-1}}(y_n).$$

Since $y_n \in K_0$, it follows that $x \in K_n$. This holds for every n, and $x \in K_0$ as well, so $x \in \bigcap_{n\geq 0} K_n$.

Second inclusion: $\bigcap_{n\geq 0} K_n \subseteq C_\rho$. Let $x \in \bigcap_{n\geq 0} K_n$. Membership in each K_n means that we can represent this same fixed number using n digits and a remainder. Choose one such representation for each n. The first three have the form

$$\begin{aligned}
x &= (1-\rho)a_0 + \rho y_1, \\
x &= (1-\rho)c_0 + (1-\rho)\rho c_1 + \rho^2 y_2, \\
x &= (1-\rho)d_0 + (1-\rho)\rho d_1 + (1-\rho)\rho^2 d_2 + \rho^3 y_3, \\
&\vdots
\end{aligned}$$

Each digit is either 0 or 1, and every y_n belongs to $[0,1]$. Different letters emphasize that representations using different numbers of digits need not initially agree in their first, second, or later digits. We now select digits that form a single infinite sequence.

First, each representation begins with either 0 or 1. At least one of these two values must occur as the first digit in infinitely many representations. Call that value b_0 and keep only those representations.

Next, discard any representation with fewer than two digits. Infinitely many remain, and their second digit is again either 0 or 1. Choose a value occurring in that position infinitely often, call it b_1, and keep only those representations. They now all begin with b_0, b_1.

Continue in the same way. At each step we discard the finitely many representations that are too short, then choose the next digit so that infinitely many representations agree with all digits chosen so far. This constructs a sequence $b_0, b_1, b_2, \ldots$.

To check that its series equals x, fix any $m \geq 1$. There are still representations using $n \geq m$ digits whose first m digits are $b_0, \ldots, b_{m-1}$. In any such representation, everything after these first m terms, including $\rho^n y_n$, is nonnegative and at most

$$(1-\rho)\sum_{j=m}^{n-1}\rho^j + \rho^n = \rho^m.$$

Therefore the partial sums of the single sequence we have chosen satisfy

$$0 \leq x - (1-\rho)\sum_{j=0}^{m-1}\rho^j b_j \leq \rho^m.$$

Since $\rho^m \to 0$, these partial sums converge to x. Hence $x = (1-\rho)\sum_{j=0}^{\infty}\rho^j b_j$, so $x \in C_\rho$. This proves the equality.

This intersection is closed and contained in $[0,1]$, hence compact by the Heine–Borel theorem. Finally, $\mathcal{K}_\rho = \{(x, 1-x) : x \in C_\rho\}$ is the image of this compact set under the continuous map $x \mapsto (x, 1-x)$, so it is compact. □

Strict advantage and persistent use. Define the maximum advantage of product 2 on the boundary set by

$$\mathcal{A}_\rho = \max_{z \in \mathcal{K}_\rho}\left\{-\Delta c + \gamma \Delta p\left[V_\rho(T_\rho^s z) - V_\rho(T_\rho^f z)\right]\right\}. \tag{4}$$

This maximum exists by continuity and compactness.

Proposition 1 (Persistent use of product 2)**.** *If $\mathcal{A}_\rho > 0$, then there is a constant $\eta > 0$, such that*

$$\liminf_{N\to\infty}\frac{1}{N}\sum_{n=1}^{N}\mathbf{1}\{i_n = 2\} \geq \eta \qquad a.s, \tag{5}$$

where i_n is the product chosen at the nth interaction with the provider A. In particular, product 2 is used infinitely often.

Proof. Fix an optimal policy and an initial state $x_0 \in \mathcal{Z}$. Throughout the proof, $x = (x_s, x_f)$ denotes the full normalized state in

$$\mathcal{Z} = \{(x_s, x_f) : x_s \geq 0,\ x_f \geq 0,\ x_s + x_f \leq 1\},$$

and $\|\cdot\|$ denotes the Euclidean norm. Let x_n be the state after the nth interaction with provider A. Thus, product i_n is chosen at state x_{n-1}.

Since D_ρ is continuous and strictly positive on the compact set $\mathcal{Z}$, it is bounded below by a positive constant. Consequently, each successive interaction with provider A occurs in finite calendar time almost surely, so the interaction sequence is well defined.

To locate states where product 2 is uniquely optimal, define, for every $x \in \mathcal{Z}$,

$$a_\rho(x) := -\Delta c + \gamma \Delta p \left[V_\rho(T_\rho^s(x)) - V_\rho(T_\rho^f(x)) \right].$$

By continuity of V_ρ and the transition maps, a_ρ is continuous on $\mathcal{Z}$. Since $\mathcal{K}_\rho$ is compact, the maximum

$$\mathcal{A}_\rho = \max_{x \in \mathcal{K}_\rho} a_\rho(x)$$

is attained. Choose $x^* \in \mathcal{K}_\rho$ such that

$$a_\rho(x^*) = \mathcal{A}_\rho > 0.$$

By continuity at x^*, there exists $\varepsilon > 0$ such that, for all $x \in \mathcal{Z}$,

$$\|x - x^*\| < \varepsilon \quad \Longrightarrow \quad |a_\rho(x) - \mathcal{A}_\rho| < \frac{\mathcal{A}_\rho}{2}.$$

In particular,

$$\|x - x^*\| < \varepsilon \quad \Longrightarrow \quad a_\rho(x) > \mathcal{A}_\rho - \frac{\mathcal{A}_\rho}{2} = \frac{\mathcal{A}_\rho}{2} > 0. \tag{6}$$

Here $a_\rho(x)$ is the difference between the values of choosing products 2 and 1, conditional on a purchase from provider A and with optimal continuation in both cases. Therefore, product 2 is uniquely optimal at every state in this neighborhood.

We next show that a single finite outcome sequence brings every starting state into this neighborhood. For any $x, \widetilde{x} \in \mathcal{Z}$, applying the success transition to both states gives

$$\begin{aligned} T_\rho^s(x) - T_\rho^s(\widetilde{x}) &= (\rho x_s + 1 - \rho, \rho x_f) - (\rho \widetilde{x}_s + 1 - \rho, \rho \widetilde{x}_f) \\ &= \rho(x_s - \widetilde{x}_s, x_f - \widetilde{x}_f) \\ &= \rho(x - \widetilde{x}). \end{aligned}$$

The same identity holds for a failure. Hence, for either $o \in \{s, f\}$,

$$\|T_\rho^o(x) - T_\rho^o(\widetilde{x})\| = \rho \|x - \widetilde{x}\|.$$

For a sequence $w = (w_1, \ldots, w_m) \in \{s, f\}^m$, listed in chronological order, write

$$T_\rho^w := T_\rho^{w_m} \circ \cdots \circ T_\rho^{w_1}.$$

Repeated application of the preceding identity yields

$$\|T_\rho^w(x) - T_\rho^w(\widetilde{x})\| = \rho^m \|x - \widetilde{x}\|. \tag{7}$$

Since the diameter of $\mathcal{Z}$ is $\sqrt{2}$, choose an integer $m \geq 1$ such that

$$\sqrt{2}\, \rho^m < \varepsilon.$$

This choice of m does not depend on the starting state.

The invariance identity

$$\mathcal{K}_\rho = T_\rho^s(\mathcal{K}_\rho) \cup T_\rho^f(\mathcal{K}_\rho),$$

iterated m times, gives

$$\mathcal{K}_\rho = \bigcup_{w\in\{s,f\}^m} T_\rho^w(\mathcal{K}_\rho).$$

Because $x^* \in \mathcal{K}_\rho$, there exist a sequence $w \in \{s,f\}^m$ and a state $\widetilde{x} \in \mathcal{K}_\rho$ such that

$$T_\rho^w(\widetilde{x}) = x^*.$$

Fix this sequence w. For every starting state $x \in \mathcal{Z}$,

$$\begin{aligned} \|T_\rho^w(x) - x^*\| &= \|T_\rho^w(x) - T_\rho^w(\widetilde{x})\| \\ &= \rho^m \|x - \widetilde{x}\| \\ &\le \sqrt{2}\,\rho^m < \varepsilon. \end{aligned} \tag{8}$$

Thus, the same sequence w brings every starting state into the neighborhood in (6).

To bound the probability of this sequence conditional on past observations, let $\mathcal{F}_n$ denote the complete history through the outcome of interaction n, including the initial state and past product choices. Set

$$q := \min\{p_1, 1 - p_2\} > 0.$$

At every interaction, regardless of the chosen product, the conditional probability of success is at least p_1, and that of failure is at least $1 - p_2$. Therefore,

$$\begin{aligned} &\Pr(\text{success at interaction } n+1 \mid \mathcal{F}_n) \ge p_1 \ge q, \\ &\Pr(\text{failure at interaction } n+1 \mid \mathcal{F}_n) \ge 1 - p_2 \ge q, \end{aligned} \qquad \text{almost surely.}$$

By successive conditioning, the conditional probability that the next m outcomes equal the fixed sequence w is at least q^m, regardless of the previous history. This does not require independence between outcomes or blocks.

Divide the interactions into consecutive, nonoverlapping blocks of length m. Block k contains interactions

$$(k-1)m + 1, \ldots, km.$$

Define the indicator

$$X_k := \mathbf{1}\{\text{the outcomes in block } k \text{ equal } w\}.$$

Then

$$\mathbb{E}[X_k \mid \mathcal{F}_{(k-1)m}] = \Pr(X_k = 1 \mid \mathcal{F}_{(k-1)m}) \ge q^m \qquad \text{almost surely.} \tag{9}$$

To turn (9) into an almost-sure frequency bound, define $M_0 = 0$ and, for $k \ge 1$,

$$M_k := \sum_{j=1}^{k} \left(X_j - \mathbb{E}[X_j \mid \mathcal{F}_{(j-1)m}]\right).$$

We verify that $(M_k)_{k\ge 0}$ is a martingale with respect to $(\mathcal{F}_{km})_{k\ge 0}$.

First, X_j is $\mathcal{F}_{jm}$-measurable and $\mathbb{E}[X_j \mid \mathcal{F}_{(j-1)m}]$ is $\mathcal{F}_{(j-1)m}$-measurable. For $j \le k$,

$$\mathcal{F}_{(j-1)m} \subseteq \mathcal{F}_{jm} \subseteq \mathcal{F}_{km}.$$

Thus, every term in M_k is determined by the history available after block k, so M_k is $\mathcal{F}_{km}$-measurable.

Second, since X_j and its conditional expectation belong to $[0,1]$,

$$\left|X_j - \mathbb{E}[X_j \mid \mathcal{F}_{(j-1)m}]\right| \le 1.$$

Consequently,

$$\mathbb{E}[|M_k|] \le \mathbb{E}\left[\sum_{j=1}^{k} \left|X_j - \mathbb{E}[X_j \mid \mathcal{F}_{(j-1)m}]\right|\right] \le k < \infty.$$

In fact, $|M_k| \le k$, so $\mathbb{E}[M_k^2] \le k^2 < \infty$ as well.

Third, separating the last increment and taking conditional expectations gives

$$\begin{aligned}\mathbb{E}[M_k \mid \mathcal{F}_{(k-1)m}] &= M_{k-1} + \mathbb{E}\big[X_k - \mathbb{E}[X_k \mid \mathcal{F}_{(k-1)m}] \mid \mathcal{F}_{(k-1)m}\big] \\ &= M_{k-1} + \mathbb{E}[X_k \mid \mathcal{F}_{(k-1)m}] - \mathbb{E}[X_k \mid \mathcal{F}_{(k-1)m}] \\ &= M_{k-1}.\end{aligned}$$

This establishes the martingale property.

Also, the tower property gives

$$\mathbb{E}\big[X_j - \mathbb{E}[X_j \mid \mathcal{F}_{(j-1)m}]\big] = \mathbb{E}[X_j] - \mathbb{E}[X_j] = 0,$$

so $\mathbb{E}[M_k] = 0$ for every k.

To obtain almost-sure convergence, observe that

$$|M_k - M_{k-1}| = \left|X_k - \mathbb{E}[X_k \mid \mathcal{F}_{(k-1)m}]\right| \le 1.$$

Hence

$$\mathbb{E}[(M_k - M_{k-1})^2] \le 1,$$

and therefore

$$\sum_{k=1}^{\infty} \frac{\mathbb{E}[(M_k - M_{k-1})^2]}{k^2} \le \sum_{k=1}^{\infty} \frac{1}{k^2} < \infty.$$

The martingale strong law, in the form of [1, Exercise 4.4.11] with $b_k = k$, now yields

$$\frac{M_k}{k} \longrightarrow 0 \qquad \text{almost surely.} \tag{10}$$

By the definition of M_k and (9),

$$\begin{aligned}\frac{1}{k}\sum_{j=1}^{k} X_j &= \frac{1}{k}\sum_{j=1}^{k} \mathbb{E}[X_j \mid \mathcal{F}_{(j-1)m}] + \frac{M_k}{k} \\ &\ge q^m + \frac{M_k}{k}.\end{aligned}$$

Taking lower limits and using (10), we obtain

$$\liminf_{k\to\infty} \frac{1}{k}\sum_{j=1}^{k} X_j \ge q^m \qquad \text{almost surely.} \tag{11}$$

It remains to relate these block occurrences to choices of product 2. If $X_j = 1$, then (8) implies

$$\|x_{jm} - x^*\| < \varepsilon.$$

By (6), product 2 is uniquely optimal at x_{jm}, and the optimal policy must therefore choose it at interaction $jm + 1$. Thus,

$$\mathbf{1}\{i_{jm+1} = 2\} \geq X_j.$$

For a horizon of N interactions, this guaranteed product-2 choice is included in the count only when

$$jm + 1 \leq N \quad \Longleftrightarrow \quad j \leq \frac{N-1}{m}.$$

Define

$$K_N := \left\lfloor \frac{N-1}{m} \right\rfloor.$$

The interactions $jm + 1$, for $1 \leq j \leq K_N$, are distinct and belong to the first N interactions. Consequently,

$$\sum_{n=1}^{N} \mathbf{1}\{i_n = 2\} \geq \sum_{j=1}^{K_N} \mathbf{1}\{i_{jm+1} = 2\} \geq \sum_{j=1}^{K_N} X_j.$$

For $N \geq m + 1$, so that $K_N \geq 1$, dividing by N gives

$$\frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\{i_n = 2\} \geq \frac{K_N}{N} \left(\frac{1}{K_N} \sum_{j=1}^{K_N} X_j \right).$$

Since

$$\frac{N-1}{mN} - \frac{1}{N} < \frac{K_N}{N} \leq \frac{N-1}{mN},$$

we have $K_N/N \to 1/m$ and $K_N \to \infty$.

The averages $K_N^{-1} \sum_{j=1}^{K_N} X_j$ belong to $[0, 1]$, so taking lower limits yields

$$\begin{aligned} \liminf_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\{i_n = 2\} &\geq \liminf_{N\to\infty} \frac{K_N}{N} \left(\frac{1}{K_N} \sum_{j=1}^{K_N} X_j \right) \\ &= \frac{1}{m} \liminf_{N\to\infty} \frac{1}{K_N} \sum_{j=1}^{K_N} X_j \\ &\geq \frac{q^m}{m} > 0 \qquad \text{almost surely,} \end{aligned}$$

where the final inequality follows from (11).

Taking $\eta = q^m/m$ proves the assertion. Since finitely many uses of product 2 would give a limiting frequency of zero, product 2 is used infinitely often almost surely. □

Proposition 2 (Characterization of positive advantage)**.** *Let U_ρ^1 be the value of always choosing product 1. For $\rho \in (0, 1)$, we have $\mathcal{A}_\rho > 0$ if and only if*

$$\max_{z \in \mathcal{K}_\rho} \left[U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z) \right] > \frac{\Delta c}{\gamma \Delta p}. \tag{12}$$

Proof. Conditional on a purchase from seller A, let

$$Q_\rho^i(z) = R - c_i + \gamma \left[p_i U_\rho^1(T_\rho^s z) + (1 - p_i) U_\rho^1(T_\rho^f z) \right], \qquad i \in \{1, 2\}.$$

This is the value of choosing product i for the current purchase and product 1 thereafter. Subtracting the two expressions gives

$$Q_\rho^2(z) - Q_\rho^1(z) = -\Delta c + \gamma \Delta p \left[U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z) \right].$$

Define

$$\mathcal{A}_\rho^1 = \max_{z \in \mathcal{K}_\rho} \left\{ -\Delta c + \gamma \Delta p \left[U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z) \right] \right\}.$$

The fixed-policy operator is a contraction preserving continuity, so U_ρ^1 is continuous and this maximum exists. We first show that $\mathcal{A}_\rho > 0$ if and only if $\mathcal{A}_\rho^1 > 0$, by proving that their nonpositivity conditions are equivalent.

Suppose $\mathcal{A}_\rho^1 \leq 0$. Then $Q_\rho^1(z) \geq Q_\rho^2(z)$ at every $z \in \mathcal{K}_\rho$. By the fixed-policy Bellman equation,

$$U_\rho^1(z) = \frac{D_\rho(z)}{1 - \gamma[1 - D_\rho(z)]} Q_\rho^1(z) = \frac{D_\rho(z)}{1 - \gamma[1 - D_\rho(z)]} \max_{i \in \{1,2\}} Q_\rho^i(z).$$

Thus U_ρ^1 also satisfies the optimal Bellman equation on $\mathcal{K}_\rho$. Since this set is invariant under both transitions, the Bellman operator restricted to it is a contraction. Uniqueness therefore gives $U_\rho^1 = V_\rho$ on $\mathcal{K}_\rho$, and hence $\mathcal{A}_\rho = \mathcal{A}_\rho^1 \leq 0$.

Conversely, suppose $\mathcal{A}_\rho \leq 0$. Product 1 then maximizes the Bellman expression evaluated at V_ρ at every state in $\mathcal{K}_\rho$. Consequently, V_ρ satisfies the Bellman equation for always choosing product 1 on this invariant set. Uniqueness of the fixed-policy value gives $V_\rho = U_\rho^1$ there, and hence $\mathcal{A}_\rho^1 = \mathcal{A}_\rho \leq 0$.

We have proved $\mathcal{A}_\rho \leq 0$ if and only if $\mathcal{A}_\rho^1 \leq 0$. Taking negations gives the desired equivalence of strict positivity. Finally, since Δc and $\gamma \Delta p > 0$ do not depend on z,

$$\mathcal{A}_\rho^1 = -\Delta c + \gamma \Delta p \max_{z \in \mathcal{K}_\rho} \left[U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z) \right].$$

Therefore $\mathcal{A}_\rho^1 > 0$ is equivalent to the inequality in the statement. □

References

[1] Rick Durrett. *Probability: Theory and Examples.* Fifth edition. Cambridge University Press, 2019.